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A Vehicle Routing Problem with Flexible Time Windows

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Abstract

In this paper, we introduce the Vehicle Routing Problem with Flexible Time Windows (VRPFlexTW), in which vehicles are allowed to deviate from customer time windows by a given tolerance. This flexibility enables savings in the operational costs of carriers, since customers may be served before and after the earliest and latest time window bounds, respectively. However, as time window deviations are undesired from a customer service perspective, a penalty proportional to these deviations is accounted for in the objective function. We develop a solution procedure, in which feasible vehicle routes are constructed via a tabu search algorithm. Furthermore, we propose a linear programming model to handle the detailed scheduling of customer visits for given routes. We validate our solution procedure by a number of Vehicle Routing Problem with Time Windows (VRPTW) benchmark instances. We highlight the costs involved in integrating flexibility in time windows and underline the advantages of the VRPFlexTW, when compared to the VRPTW.

Keywords: Vehicle routing problems, Time windows, Soft time windows, Flexible time windows

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1. Introduction

Carrier companies are faced with the daily challenge of delivering goods to customers in a cost-effective manner. Often times, these companies must adhere to customer service requirements. In this environment, customer service requirements are mainly reflected by the Vehicle Routing Problem with Time Windows (VRPTW). This problem can be observed in bank deliveries, postal deliveries and school bus routing (see Hashimoto et al. [14]). Given a set of customers, the VRPTW consists of finding least cost routes such that each customer is visited within a predetermined time window by a single vehicle. Furthermore, a vehicle must deliver a quantity not exceeding its capacity, the vehicle should also start and end its route at a given depot. The vehicle is permitted to arrive before the opening of the time window, and wait at no cost until service becomes possible, but it is not permitted to arrive after the time window closes (see Bräysy and Gendreau [4]).

The definition of the VRPTW implies that time windows are treated as hard constraints, the relaxation of which may lead to reducing the total travel time while using fewer vehicles. A form of time window relaxation is considered in the Vehicle Routing Problem with Soft Time Windows (VRPSTW). This problem assumes that some or all customer time windows are soft and can be violated by paying appropriate penalties (see Balakrishnan [1]). The penalty structure associated with soft time windows essentially allows serving a customer at any point of the planning horizon. This mechanism is due to the penalty policies, which dictate that early arriving vehicles must wait or incur a penalty, while any late arrival is permissible at a cost. Therefore, when compared to the VRPTW, the VRPSTW operates on a much larger feasible solution space.

In several real-world situations, time window constraints can be violated to a certain extent. Therefore, in this paper we aim to assess the operational gains obtained by employing a fixed relaxation of the time window constraints. Namely, we study the Vehicle Routing Problem with Flexible Time Windows (VRPFlexTW), in which vehicles are allowed to deviate from customer time windows by a given tolerance. As the time window violations affect customers satisfaction, they are penalized. Furthermore, as in the VRPTW we allow early arriving vehicles to wait at no cost until the earliest allowable service time is reached. The VRPFlexTW is distinct from the VRPSTW in that the former considers a restriction on the feasible time window violation. Therefore, when compared to the VRPSTW, the

VRPFlexTW operates on a smaller feasible solution space.

The main contributions of this paper are threefold:

1. We introduce and model the VRPFlexTW.
2. We develop a solution procedure, which comprises three phases, for producing high-quality solutions.
3. We conduct a series of numerical experiments on benchmark instances, and assess the operational gains of using flexible time windows.

The remainder of this paper is organized as follows. The relevant literature is reviewed in Section 2. Section 3 introduces the model. The solution procedure is then described in Section 4. This is followed by computational results provided in Section 5 and by conclusions given in Section 6.

2. Literature review

The daily distribution task faced by many freight transports is captured by the Vehicle Routing Problem (VRP). In its classical definition, the VRP minimizes the total travel cost incurred by a set of homogeneous vehicles that deliver customer demands. Each customer is to be visited by a single vehicle, each vehicle starts and ends its route at a depot and delivers a quantity not exceeding its capacity. The VRP has been widely studied for over fifty years (see, e.g., Laporte [19]). In an attempt to better link the VRP to realistic applications, a number of extensions have been proposed in the literature (see, e.g., Toth and Vigo [27], and Golden et al. [13]). One of the most extensively studied variants of the VRP is the VRPTW, in which time windows ensure that a customer must be visited within a given interval. Over the years, a number of exact and heuristic solution procedures have been proposed for the VRPTW. Bräysy and Gendreau [4] review the literature on route construction and local search algorithms for the VRPTW. The authors also survey metaheuristics for the VRPTW (see Bräysy and Gendreau [5]). Baldacci et al. [3] provide a recent review of mathematical formulations, relaxations and exact methods for the VRPTW.

The VRPTW treats time windows as hard constraints. However, some practical applications imply that customer time windows could be treated as soft constraints, i.e., may be violated at a cost. This setting gives rise to the VRPSTW, which is significantly less studied than the VRPTW. The majority of the literature on the VRPSTW considers a linear penalty function for time window deviations. Balakrishnan [1] develops three heuristics

for the VRPSTW based on the nearest neighbor Clarke-Wright savings and space-time rules. Koskosidis et al. [18] propose a heuristic algorithm for the VRPSTW. Their algorithm decomposes the problem into an assignment component and a series of routing and scheduling components. Min [20] considers the VRPSTW for a single vehicle where the problem is solved for small instances. Taillard et al. [25] propose a tabu search heuristic to solve the VRPSTW.

Ibaraki et al. [16] propose an efficient algorithm to deal with general time window constraints. The cost function considered for time window violations can be non-convex and discontinuous as long as it is piecewise linear. Furthermore, one or more time slots can be assigned to each customer. Building upon the model proposed in [16], Hashimoto et al. [14] introduce traveling times as variables. In the latter study, the traveling time represents the difference between the starting times of services at two consecutive customers and is introduced in the cost function.

Calvete et al. [6] consider a general medium-sized VRPSTW and propose a goal programming model. Aside from minimizing the operational cost and time window violations, the authors consider avoiding underutilization of vehicles and labor. The solution approach first computes feasible routes and then selects the set of best ones.

One of the underlying assumptions in VRPSTW is that the deviations from the time windows are essentially unbounded, implying that any feasible VRP solution is feasible in the VRPSTW as well. The VRPFlexTW proposed in this paper bounds the lower and upper time window deviations, and hence allows a predetermined amount of flexibility in adhering to time windows. Qureshi et al. [21, 22] develop a column generation based exact algorithm for the Vehicle Routing and scheduling Problem with Semi Soft Time Windows (VRPSSTW). This problem considers an upper bound on the tardiness time window deviation, and thus may be viewed as a special case of the VRPFlexTW. The solution approach is shown to be efficient on medium-sized instances. Tang et al. [26] study the VRP with fuzzy time windows where the authors consider the multi-objective problem of minimizing travel time and maximizing customer service level, similar to the VRPTW. The authors take into account a limited allowable deviation from time windows and solve their multi-objective model with a two-stage algorithm which yields Pareto solutions.

3. Model Formulation

Formally, the VRPFlexTW can be represented by a connected digraph $G = (N, A)$ where $N = \{0, 1, ..., n\}$ is the set of nodes and $A = \{(i, j)|i, j \in N, i \neq j\}$ is the set of arcs. The node 0 corresponds to the central depot and the other nodes in N denote customers that must be served. For each customer $i \in N \setminus \{0\}$, we have a positive demand q_i, a time window $[l_i, u_i]$ and a percentage p_i which is used to set the maximum allowed violation, leading to the flexible time window. The time window at the depot, $[l_0, u_0]$, corresponds to the feasible scheduling horizon for each vehicle route. For each node i, a flexible time window $[l'_i, u'_i]$ is generated with respect to the length of the original time window, where $l'_i = l_i - p_i(u_i - l_i)$, and $u'_i = u_i + p_i(u_i - l_i)$. Additionally, Q represents the capacity given for each vehicle $v \in V$ where V denotes a homogeneous fleet.

Associated with each arc $(i, j) \in A$, t_{ij} and d_{ij} represent the travel time and the distance along that arc, respectively. Note that the service time at node i, z_i is included by t_{ij}. A fixed cost c_f is incurred for using a vehicle. Time window violations, i.e., serving a customer within $[l'_i, l_i]$ or $[u_i, u'_i]$ are penalized by c_e and c_d for one unit of earliness and one unit of delay, respectively. Moreover, c_t is the cost paid for one unit of distance. In the early servicing case, service at the customer starts between the flexible earliest time and the original earliest time. In the late servicing case, service takes place between the original latest time and the flexible latest time. Note that vehicles wait at customers (at least) until the flexible time window is reached if they arrive early, and they cannot serve after the customer flexible time window closes. In our problem definition, we assume that waiting brings no penalty cost. The latter assumption enables vehicles to wait at customer locations even if they arrive within the flexible time windows, and to generate cost-efficient routes. Figure 1 depicts the possible arrivals and their corresponding penalty cases at customers.

Under these assumptions, the mathematical model is formulated as follows.

$$\begin{aligned} \min \quad & c_t \sum_{i \in N} \sum_{j \in N} \sum_{v \in V} d_{ij} x_{ijv} + c_f \sum_{j \in N \setminus \{0\}} \sum_{v \in V} x_{0jv} \\ & + c_e \sum_{i \in N} \sum_{v \in V} e_{iv} + c_d \sum_{i \in N} \sum_{v \in V} h_{iv} \end{aligned} \tag{1}$$

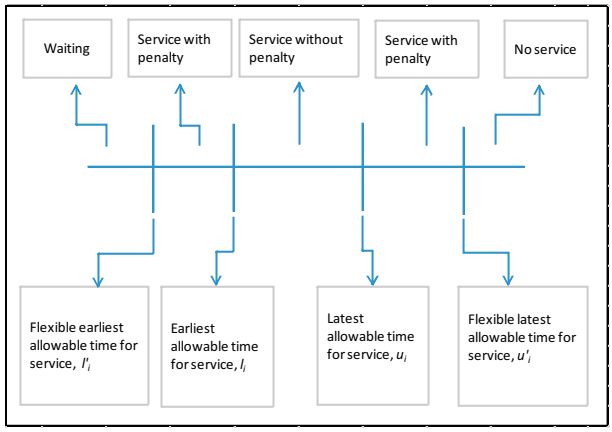


Figure 1: Possible arrivals and their corresponding penalty cases at customers

$$\text{subject to} \quad \sum_{j\in N}\sum_{v\in V} x_{ijv} = 1, \qquad i \in N \setminus \{0\}, \tag{2}$$

$$\sum_{i\in N} x_{ikv} - \sum_{j\in N} x_{kjv} = 0, \qquad k \in N \setminus \{0\}, v \in V, \tag{3}$$

$$\sum_{i\in N\setminus\{0\}} q_i \sum_{j\in N} x_{ijv} \le Q, \qquad v \in V, \tag{4}$$

$$\sum_{j\in N} x_{0jv} = 1, \qquad v \in V, \tag{5}$$

$$\sum_{i\in N} x_{i0v} = 1, \qquad v \in V, \tag{6}$$

$$s_{iv} + t_{ij} - M(1 - x_{ijv}) \le s_{jv}, \qquad i \in N, j \in N, v \in V, \tag{7}$$

$$l'_i \le s_{iv} \le u'_i, \qquad i \in N, v \in V, \tag{8}$$

$$e_{iv} \ge 0, \qquad i \in N, v \in V, \tag{9}$$

$$e_{iv} \ge l_i - s_{iv}, \qquad i \in N, v \in V, \tag{10}$$

$$h_{iv} \ge 0, \qquad i \in N, v \in V, \tag{11}$$

$$h_{iv} \ge s_{iv} - u_i, \qquad i \in N, v \in V, \tag{12}$$

$$x_{ijv} \in \{0, 1\}, \qquad i \in N, j \in N, v \in V. \tag{13}$$

In the above model, x_{ijv} is equal to 1 if vehicle v serves node j immediately after node i and 0, otherwise. s_{iv} denotes the time that vehicle v starts serving node i. Furthermore, e_{iv} and h_{iv} represent the earliness and the delay

at node i in case it is served by vehicle v, respectively. The objective (1) is to minimize the total cost which consists of traveling costs, fixed costs of vehicles used for service, and penalty costs incurred for early and late servicing. The constraints (2) and (3) guarantee that exactly one vehicle arrives at a customer location and leaves it. The constraints (4) ensure that the vehicle capacity is not exceeded. The constraints (5) and (6) indicate that each vehicle route starts and terminates at the depot. The constraints (7) represent the relationship between the starting time of service at a customer and the departure time of vehicle from its predecessor. The constraints (8) ensure that the service takes place at each customer with respect to the customer (flexible) time window. The constraints (9) and (10) link the earliness and the beginning of service; similarly, the constraints (11) and (12) link the delay and the beginning of service. The constraints (13) indicate that there is no partial servicing.

4. Solution Methodology

For the VRPFlexTW formally described above, we propose a solution method that includes three main phases. In the first phase, an initial feasible solution is constructed. This solution is then improved by applying a tabu search metaheuristic in the second phase. These two procedures lead to the assignments of vehicles and the sequences of customers in these assignments. In the third phase, the solution obtained by the tabu search algorithm is further improved by solving a Linear Programming (LP) model. This phase calculates the optimal starting time of each vehicle route from the depot, and optimal times that each vehicle should start serving the customers in its route. The objective function of the LP minimizes the total penalty cost of the vehicle route with respect to the sequence of customers given in that route. In the first and second phases, vehicles are allowed to wait at customer locations only in case they arrive early (until the flexible time window is reached). If a vehicle arrives at a customer within its flexible time window, then service takes place without waiting. The latter situation leads to the immediate service with some penalty costs.

4.1. Initial Feasible Solution

We apply the time-oriented nearest neighbor heuristic proposed by Solomon [23] to generate our initial routes. Suppose that we have a partial route for vehicle v in which the last demand location is node i, and node j is any node

that can be visited by that vehicle after node i. Following Solomon [23], we use a function to calculate the cost value of node j in case it is visited by vehicle v immediately after node i. This function uses three measures to evaluate the insertion of node j with respect to the last node in the partial route. The first measure is the distance between node i and node j. The other two measures are formally given as follows:

$$T_{jv} = s_{jv} - (s_{iv} + z_i), \text{ and} \tag{14}$$

$$r_{jv} = u'_j - (s_{iv} + t_{ij}). \tag{15}$$

s_{jv}, which is the time that vehicle v starts serving customer j, is calculated by:

$$s_{jv} = max\{l'_j, s_{iv} + t_{ij}\}. \tag{16}$$

The cost value is then calculated by the following equation:

$$c_{jv} = \beta_1 d_{ij} + \beta_2 T_{jv} + \beta_3 r_{jv}, \tag{17}$$

where $\beta_1 \geq 0$, $\beta_2 \geq 0$, $\beta_3 \geq 0$ and $\beta_1 + \beta_2 + \beta_3 = 1$.

The above cost calculation is operated for a set of unrouted customers, which can feasibly be inserted to the current partial route, to choose the customer that has the minimum insertion cost value. The feasibility is checked with respect to the flexible time window at the customer considered, the flexible time window at the depot and the capacity of the vehicle. If we cannot find any feasible customer for the current vehicle route, a new partial route is then constructed by inserting the unrouted customer which has the minimum cost value with respect to the depot. This procedure terminates routing when all customers are assigned to a vehicle.

4.2. The Tabu Search for the VRPFlexTW

The tabu search metaheuristic has been extensively applied to the classical VRP and its extensions, such as stochastic VRP and VRPTW. Some tabu search methods proposed for the VRP are given by Gendreau et al. [11, 12]. The interested reader is referred to Garcia et al. [10], Taillard et al. [25], Hertz et al. [15] and Cordeau et al. [8] for implementations of tabu search in the VRPTW. Our tabu search algorithm is based on the work of Taş et al. [24]. In this study, the authors focus on a VRP with stochastic travel times and soft time windows. Travel times on each arc are given with a known probability distribution. Soft time windows allow both early and late

servicing with some penalty costs. The latter enables vehicles to start serving a customer after its time window closes. Moreover, vehicles do no wait at customer locations, leading to immediate service regardless of the arrival time. We adapt this algorithm to our problem where we have deterministic travel times and flexible time windows. Each time window is relaxed with a given percentage for each customer. The flexible time boundaries can be thought as the hard time windows in the classical VRPTW where vehicles are not permitted to serve before or after these intervals.

The overall tabu search procedure is described in pseudo-code as Algorithm 1. In this algorithm, y and $z(y)$ denote the current solution and its total cost value, respectively. The latter value is obtained by Algorithm 2 which calculates the total cost value of each route k in solution y. In this algorithm, penalties incurred due to violations of the original time windows are computed with respect to the Earliest Possible beginning of Service (EPS) heuristic. Note that in the tabu search method, solution y is taken into consideration in case all routes in this solution are feasible with respect to the flexible time window constraints.

The neighborhood of the current solution y, which is represented by $g(y)$, is generated by two types of operators: (i) changing the location of the customer within the route, (ii) removing the customer from a route and locating it into another route. Each solution y' in $g(y)$ is evaluated with respect to $c(y')$, which is calculated as follows:

$$c(y') = z(y') + \nu q(y'), \tag{18}$$

where $q(y')$ is the total demand of routes in solution y' exceeding the vehicle capacity and ν is the cost paid for one unit of excess demand. This calculation is operated in line with Cordeau et al. [8] and Taş et al. [24]. The parameter ν is adjusted after each iteration with respect to the total demand of routes in the current solution. If the current solution is feasible with respect to the vehicle capacity constraint, the value of ν is divided by $(1 + \varphi)$; otherwise it is multiplied with $(1 + \varphi)$.

In equation (18), an additional cost is added to $c(y')$ of any solution y' such that $c(y') \geq c(y)$. This mechanism provides diversification during the search. In our paper, a similar function to that given in [24] is applied to calculate additional costs. Note that the latter function uses a constant parameter (μ) to calibrate the intensity of the diversification.

```
Construct initial feasible solution, y_init
Set y := y_init, y* := y_init and z(y*) := z(y_init)
Set κ := 1, stop := 0
while κ ≤ θ and stop = 0 do
    Generate the neighborhood of y, g(y)
    Choose the first solution y' ∈ g(y) which is better than y and is
    not tabu or satisfies the aspiration criterion
    if Such a solution cannot be found then
        Choose a solution y' ∈ g(y) that minimizes c(y') value and is
        not tabu
    end
    if y' is feasible and z(y') < z(y*) then
        Set y* := y' and z(y*) := z(y')
    end
    if y* is not updated for √κ iterations then
        Set y := y* and c(y) := c(y*)
        Update the tabu list accordingly
    end
    else
        Set y := y' and c(y) := c(y')
    end
    Update ν with respect to y
    if y* is not updated for τ iterations then
        Set stop := 1
    end
    Set κ := κ + 1
end
return y*
```

Algorithm 1: The tabu search algorithm for the VRPFlexTW

In Algorithm 1, we apply two criteria following Taş et al. [24] where the algorithm terminates either because it reaches the maximum number of given tabu search iterations (θ) or because the best feasible solution did not change for a threshold number of iterations (τ). To identify the solutions that are tabu, we employ a list which includes the customers forbidden to relocate for a number of iterations. Note that the size of this list is denoted by ϑ. The interested reader is referred to [24] for the details about medium-term memory applied in Algorithm 1, which is based on directing the search to the promising regions of the neighborhood generated by the best feasible

solution.

foreach *route k in solution y* **do**
  foreach *customer j in route k* **do**
    Calculate the arrival time a_{jv} with respect to previous customer i, where $a_{jv} = s_{iv} + t_{iv}$
    if $a_{jv} < l'_j$ **then**
      Set $s_{jv} := l'_j$ and calculate earliness
    end
    if $l'_j \leq a_{jv} < l_j$ **then**
      Set $s_{jv} := a_{jv}$ and calculate earliness
    end
    if $l_j \leq a_{jv} < u_j$ **then**
      Set $s_{jv} := a_{jv}$ (no penalty)
    end
    if $u_j \leq a_{jv} < u'_j$ **then**
      Set $s_{jv} := a_{jv}$ and calculate delay
    end
  end
  Calculate the total cost of route k by using its total distance, total earliness and total delay, and the cost of vehicle v
end
Calculate the total cost of solution y, $z(y)$

Algorithm 2: EPS heuristic to calculate $z(y)$

In Algorithm 2, v denotes the vehicle that operates route k. Moreover, a_{jv} and s_{jv} represent the arrival time and the beginning of service of vehicle v at node j, respectively.

4.3. Scheduling Method

We solve the following LP model to obtain the optimal times that each vehicle starts serving the customers in its route. This model is operated for each vehicle route generated by the tabu search algorithm.

$$\min \quad c_e \sum_{i \in N} e_{iv} + c_d \sum_{i \in N} h_{iv} \tag{19}$$

$$\text{subject to} \quad s_{iv} + t_{ij} \leq s_{jv}, \qquad i \in N^{'}, j \in N^{'}, (i,j) \in A^{'}, \tag{20}$$

$$l_i' \leq s_{iv} \leq u_i', \qquad i \in N^{'}, \tag{21}$$

$$e_{iv} \geq 0, \qquad i \in N^{'}, \tag{22}$$

$$e_{iv} \geq l_i - s_{iv}, \qquad i \in N^{'}, \tag{23}$$

$$h_{iv} \geq 0, \qquad i \in N^{'}, \tag{24}$$

$$h_{iv} \geq s_{iv} - u_i, \qquad i \in N^{'}, \tag{25}$$

where the objective is to minimize the total penalty cost of the route operated by vehicle v. $A^{'} \subseteq A$ is the set of arcs traversed by vehicle v and $N^{'} \subseteq N$ is the set of nodes visited in the route of that vehicle.

5. Numerical Results and Insights

We perform our computational experiments on the well-known data sets given by Solomon [23]. We consider 29 problem instances with 100 customers and tight time windows (sets R1 , C1 and RC1). Each instance has one depot as the central location of the homogeneous fleet of vehicles, where the vehicle capacity Q is 200 units. Cost coefficients (c_t, c_f, c_e, c_d) are equal to (2.0, 400, 0.5, 1.0), and p_i is set to 0.05 for each node $i \in N$.

Following [23], we use the parameters given in Table 1 to generate the Initial Feasible Solution (IFS) for each problem instance. Among four solutions constructed by the initialization algorithm, we select the solution with the minimum total cost value calculated with respect to the EPS heuristic, and we set IFS as the solution selected. The algorithms proposed in our solution procedure are coded in JAVA and the LP model is solved by using IBM ILOG CPLEX 12.2 [17]. All experiments are conducted on an Intel Core Duo with 2.93 GHz and 4 GB of RAM.

Table 1: Parameters used by the initialization algorithm to generate IFS

Tests	β_1	β_2	β_3
Test 1	0.4	0.4	0.2
Test 2	0.0	1.0	0.0
Test 3	0.5	0.5	0.0
Test 4	0.3	0.3	0.4

5.1. Sensitivity Analyses and Parameters for the Tabu Search

A number of tests are performed to tune the parameters employed in the tabu search algorithm. We apply a similar procedure to that given in Cordeau et al. [7] and applied in Taş et al. [24]. To determine the most appropriate value of a parameter, we test its different values over an interval by keeping the other parameters unchanged. In our preliminary experiments, three main sets of results are obtained where the parameters μ, ϑ and φ are examined in [0.005,0.025], [5log$_{10}|N|$,15log$_{10}|N|$] and [0.25,1.25], respectively. According to results of these preliminary tests, we set the values of μ, ϑ and φ to 0.020, 5log$_{10}|N|$, and 0.50, respectively.

In the tabu search algorithm, we adjust the value of the parameter ν at each iteration. As a reasonable starting value for the penalty cost incurred due to one unit of capacity violation, the initial value of the parameter ν is set to 1.

We obtain three sets of results by applying three different sets of values for stopping criteria, in which (θ,τ) are set to (10^4,10^3), (10^4,5(10^3)) and (10^5,10^4). In the next subsection, we present our results and discuss their aspects in detail.

5.2. Results on (θ,τ)

Table 2 provides the solutions generated by the initialization algorithm for each problem instance. Note that for each starting solution, total delay (Del.), total distance (Dist.), total earliness (Earl.), number of vehicles activated for the service (#Veh.), and objective function value (Obj.) are presented. Since the computational time spent by the initialization algorithm is next to 0 for each instance, we do not report this value in Table 2.

Tables 3, 4 and 5 show the solutions obtained by the tabu search algorithm, and the corresponding final solutions obtained by solving an LP model in the scheduling method. In these tables, values of (θ,τ) are equal to (10^4,10^3), (10^4,5(10^3)) and (10^5,10^4), respectively. For the tabu search algorithm, we report the CPU times in seconds and the improvement in total cost values in percentages, which is calculated with respect to the IFS. For the LP model, improvement in total penalty cost incurred for early and late services is given in percentages and this value is calculated with respect to the solution obtained by the tabu search algorithm. Moreover, we present the averages of improvements (Avg.) over all problem instances provided both by the tabu search algorithm and by the scheduling method.

Table 2: Details of initial feasible solutions obtained for all problem instances where $p_i = 0.05$, $\forall i \in N$

	Initial Feasible Solution				
Ins.	Del.	Dist.	Earl.	#Veh.	Obj.
c101	0.00	855.07	7.30	10	5713.78
c102	35.39	1263.54	45.70	11	6985.31
c103	159.93	1530.61	96.20	11	7669.26
c104	226.46	1789.19	45.00	11	8227.33
c105	0.00	934.36	0.00	10	5868.71
c106	6.32	1164.45	0.00	10	6335.23
c107	14.72	1033.36	11.52	10	6087.20
c108	2.60	1038.91	4.20	10	6082.52
c109	10.31	1357.11	167.55	11	7208.30
r101	0.09	1948.05	60.20	19	11526.29
r102	63.01	1816.10	42.30	19	11316.36
r103	135.35	1601.99	29.00	14	8953.83
r104	121.20	1387.88	14.00	12	7703.96
r105	0.15	1706.48	75.03	17	10250.64
r106	64.38	1736.24	30.66	14	9152.19
r107	119.97	1464.20	45.53	12	7871.13
r108	86.87	1301.18	33.50	11	7105.97
r109	9.47	1487.61	90.35	13	8229.86
r110	29.17	1470.03	46.76	13	8192.61
r111	72.83	1500.50	68.65	12	7908.16
r112	55.27	1390.12	36.44	11	7253.73
rc101	1.65	2020.20	73.37	17	10878.73
rc102	66.69	1891.19	54.00	14	9476.07
rc103	77.29	1803.36	43.20	13	8905.62
rc104	89.29	1735.92	39.75	12	8381.01
rc105	12.18	2081.24	61.75	16	10605.55
rc106	5.50	1664.48	45.34	13	8557.13
rc107	20.88	1610.98	83.95	12	8084.82
rc108	17.62	1507.50	35.73	12	7850.48

Solutions in Table 3 show that in all problem instances the tabu search algorithm decreases the total distance with respect to the initial solutions. These reductions may be achieved by an increase in the delay or in the earliness, e.g., we have higher total delay in three instances (r101, r105, rc101) and higher total earliness in nine instances (r101, r102, r106, r110, rc101, rc102, rc106, rc107, rc108) compared to that given by IFS. Moreover, we use fewer vehicles in 12 problem instances. Overall, the tabu search algorithm reduces the total cost by 11.59% on average. The solutions obtained by the tabu search algorithm are further improved by the LP model by reducing an average of 60.03% of the total penalty cost.

Solutions in Table 4, which are obtained with a higher number of iterations given for the secondary terminating criterion, show that in all problem instances the tabu search algorithm decreases the total distance with respect to the initial solutions. We have higher total delay in three instances and higher total earliness in ten instances. Moreover, we use fewer vehicles in 16 instances. Overall, the tabu search algorithm reduces the total cost by 12.88% on average, which is higher than the average value generated with a smaller threshold iterations to terminate the algorithm (τ=1000). The solutions obtained by the tabu search algorithm are further improved by scheduling method by 60.88% on average according to total penalty cost.

In the last set of results, the tabu search algorithm operates with higher numbers of iterations both for the primary (θ) and the secondary (τ) terminating criteria. In all problem instances, the tabu search algorithm decreases the total distance with respect to the initial solutions. We have higher total delay in three instances and higher total earliness in ten instances. Moreover, we use fewer vehicles in 19 instances. Overall, the tabu search algorithm reduces the total cost by 14.15% on average, which is higher than the average value given in Table 4. Scheduling method yields a 58.62% improvement on average, which is slightly smaller than the value given in Table 4, since the tabu search algorithm leads to better improvements for most of the problem instances.

Table 3: Details of solutions obtained by the tabu search algorithm and the scheduling method, where (θ,τ) are (10^4,10^3) and $p_i = 0.05$, $\forall i \in N$

	Solution of the tabu search algorithm							Final Solution		
Ins.	Del.	Dist.	Earl.	#Veh.	Obj.	CPU	Imp %	Del.	Earl.	Imp %
c101	0.00	828.94	0.00	10	5657.87	51	0.98	0.00	0.00	0.00
c102	0.00	828.94	0.00	10	5657.87	92	19.00	0.00	0.00	0.00
c103	1.53	915.19	20.95	10	5842.38	209	23.82	1.53	0.00	87.22
c104	0.00	940.74	22.60	11	6292.77	182	23.51	0.00	0.00	100.00
c105	0.00	828.94	0.00	10	5657.87	76	3.59	0.00	0.00	0.00
c106	0.00	828.94	0.00	10	5657.87	91	10.69	0.00	0.00	0.00
c107	0.00	828.94	0.00	10	5657.87	87	7.05	0.00	0.00	0.00
c108	0.00	828.94	0.00	10	5657.87	78	6.98	0.00	0.00	0.00
c109	0.00	828.94	0.00	10	5657.87	84	21.51	0.00	0.00	0.00
r101	0.36	1667.80	60.68	19	10966.31	149	4.86	0.36	1.10	97.03
r102	0.28	1480.26	44.30	17	9782.94	292	13.55	0.28	0.54	97.56
r103	0.00	1293.31	24.27	14	8198.75	113	8.43	0.00	0.00	100.00
r104	0.92	1067.69	13.45	12	6943.02	79	9.88	0.92	0.00	87.97
r105	1.22	1409.07	73.14	15	8855.93	126	13.61	1.22	1.96	94.17
r106	2.23	1320.33	48.00	13	7866.89	197	14.04	2.23	6.02	80.04
r107	8.11	1093.54	40.96	12	7015.68	183	10.87	8.11	8.21	57.28
r108	5.85	987.49	30.00	11	6395.83	173	9.99	5.85	3.00	64.76
r109	2.65	1213.19	83.33	13	7670.69	168	6.79	3.98	8.92	80.95
r110	6.72	1109.93	81.63	13	7467.40	140	8.85	6.72	7.85	77.61
r111	6.74	1105.85	63.21	12	7050.03	179	10.85	6.74	12.72	65.84
r112	3.89	1034.33	26.98	10	6086.04	138	16.10	3.89	4.94	63.39
rc101	2.79	1728.53	92.18	16	9905.94	145	8.94	2.79	6.51	87.63
rc102	7.09	1530.03	72.18	14	8703.23	203	8.16	7.35	8.74	72.85
rc103	10.52	1330.93	39.94	12	7492.36	100	15.87	10.52	4.50	58.12
rc104	5.45	1208.84	31.31	11	6838.79	137	18.40	5.45	6.61	58.52
rc105	3.03	1576.17	61.64	15	9186.19	317	13.38	3.03	8.53	78.44
rc106	2.85	1401.97	74.79	13	8044.19	160	5.99	2.85	7.71	83.34
rc107	4.18	1292.44	103.08	11	7040.60	158	12.92	4.18	15.28	78.79
rc108	9.43	1205.28	68.38	12	7254.18	87	7.60	9.43	7.99	69.23
Avg.							11.59			60.03

Table 4: Details of solutions obtained by the tabu search algorithm and the scheduling method, where (θ,τ) are (10^4,5(10^3)) and $p_i = 0.05$, $\forall i \in N$

	Solution of the tabu search algorithm							Final Solution		
Ins.	Del.	Dist.	Earl.	#Veh.	Obj.	CPU	Imp %	Del.	Earl.	Imp %
c101	0.00	828.94	0.00	10	5657.87	148	0.98	0.00	0.00	0.00
c102	0.00	828.94	0.00	10	5657.87	308	19.00	0.00	0.00	0.00
c103	1.53	888.81	10.55	10	5784.42	466	24.58	1.53	0.00	77.47
c104	0.00	931.23	22.60	11	6273.77	397	23.74	0.00	0.00	100.00
c105	0.00	828.94	0.00	10	5657.87	343	3.59	0.00	0.00	0.00
c106	0.00	828.94	0.00	10	5657.87	372	10.69	0.00	0.00	0.00
c107	0.00	828.94	0.00	10	5657.87	334	7.05	0.00	0.00	0.00
c108	0.00	828.94	0.00	10	5657.87	304	6.98	0.00	0.00	0.00
c109	0.00	828.94	0.00	10	5657.87	285	21.51	0.00	0.00	0.00
r101	0.36	1661.10	63.20	19	10954.16	581	4.96	0.36	1.02	97.28
r102	0.52	1469.78	45.30	17	9762.72	598	13.73	0.52	0.88	95.87
r103	0.42	1280.21	25.00	14	8173.33	490	8.72	0.42	0.16	96.15
r104	0.83	1050.24	15.00	12	6908.80	489	10.32	0.83	0.00	90.06
r105	2.74	1373.56	89.75	15	8794.74	575	14.20	3.49	5.20	87.21
r106	2.56	1317.44	51.00	13	7862.94	492	14.09	2.56	7.52	77.47
r107	7.30	1075.55	40.60	12	6978.69	507	11.34	7.30	7.81	59.41
r108	6.34	954.93	18.00	10	5925.19	514	16.62	6.34	3.00	48.90
r109	2.65	1213.19	83.33	13	7670.69	391	6.79	3.98	8.92	80.95
r110	6.72	1106.47	69.88	12	7054.59	507	13.89	6.72	10.85	70.85
r111	5.07	1078.75	53.84	12	6989.50	491	11.62	5.07	3.23	79.09
r112	1.70	978.87	27.00	10	5972.95	427	17.66	1.70	3.56	77.13
rc101	3.91	1653.78	71.52	15	9347.23	576	14.08	3.91	5.62	83.06
rc102	2.36	1507.97	58.08	14	8647.35	567	8.75	2.83	4.08	84.49
rc103	9.09	1288.82	48.00	12	7410.74	469	16.79	9.09	4.50	65.72
rc104	2.19	1186.93	30.32	11	6791.22	494	18.97	2.19	3.00	78.74
rc105	3.03	1576.17	61.64	15	9186.19	519	13.38	3.03	8.53	78.44
rc106	3.13	1363.55	59.47	12	7559.98	520	11.65	3.13	7.56	78.96
rc107	8.24	1264.87	90.66	11	6983.31	521	13.62	8.24	17.02	68.74
rc108	2.39	1145.46	70.86	11	6728.73	449	14.29	2.39	3.24	89.41
Avg.							12.88			60.88

Table 5: Details of solutions obtained by the tabu search algorithm and the scheduling method, where (θ,τ) are (10^5,10^4) and $p_i = 0.05$, $\forall i \in N$

	Solution of the tabu search algorithm							Final Solution		
Ins.	Del.	Dist.	Earl.	#Veh.	Obj.	CPU	Imp %	Del.	Earl.	Imp %
c101	0.00	828.94	0.00	10	5657.87	225	0.98	0.00	0.00	0.00
c102	0.00	828.94	0.00	10	5657.87	577	19.00	0.00	0.00	0.00
c103	1.53	888.81	10.55	10	5784.42	821	24.58	1.53	0.00	77.47
c104	0.00	873.83	27.30	10	5761.32	1007	29.97	0.00	0.00	100.00
c105	0.00	828.94	0.00	10	5657.87	675	3.59	0.00	0.00	0.00
c106	0.00	828.94	0.00	10	5657.87	722	10.69	0.00	0.00	0.00
c107	0.00	828.94	0.00	10	5657.87	614	7.05	0.00	0.00	0.00
c108	0.00	828.94	0.00	10	5657.87	553	6.98	0.00	0.00	0.00
c109	0.00	828.94	0.00	10	5657.87	522	21.51	0.00	0.00	0.00
r101	0.36	1661.10	63.20	19	10954.16	890	4.96	0.36	1.02	97.28
r102	0.52	1469.78	45.30	17	9762.72	878	13.73	0.52	0.88	95.87
r103	0.53	1264.75	26.00	14	8143.03	1991	9.06	0.53	0.98	92.45
r104	3.72	1017.38	11.59	10	6044.28	2526	21.54	3.72	1.00	55.68
r105	2.74	1373.56	89.75	15	8794.74	867	14.20	3.49	5.20	87.21
r106	2.10	1305.30	51.85	13	7838.62	1117	14.35	2.10	6.02	81.77
r107	7.30	1075.55	40.60	12	6978.69	801	11.34	7.30	7.81	59.41
r108	6.34	954.93	18.00	10	5925.19	905	16.62	6.34	3.00	48.90
r109	1.97	1153.50	65.30	12	7141.62	2901	13.22	1.97	7.90	82.90
r110	6.72	1106.47	69.88	12	7054.59	821	13.89	6.72	10.85	70.85
r111	0.00	1079.93	53.84	12	6986.77	1097	11.65	0.00	3.23	94.00
r112	1.93	974.32	26.35	10	5963.75	1172	17.78	1.93	3.56	75.45
rc101	5.74	1641.65	89.18	15	9333.63	1660	14.20	5.74	10.12	78.54
rc102	1.53	1503.75	64.08	14	8641.08	1490	8.81	2.00	4.63	87.15
rc103	15.18	1283.40	45.00	12	7404.47	1834	16.86	15.18	4.50	53.74
rc104	5.63	1143.93	24.32	10	6305.65	2664	24.76	5.63	3.80	57.65
rc105	3.03	1576.17	61.64	15	9186.19	789	13.38	3.03	8.53	78.44
rc106	3.13	1363.55	59.47	12	7559.98	807	11.65	3.13	7.56	78.96
rc107	8.24	1264.87	90.66	11	6983.31	796	13.62	8.24	17.02	68.74
rc108	2.18	1115.90	58.98	10	6263.47	1575	20.22	2.18	9.94	77.44
Avg.							14.15			58.62

5.3. VRPFlexTW versus VRPTW

The aim of this subsection is to evaluate the benefits gained by flexible time windows compared to the hard time windows. Table 6 provides the optimal/best-known solutions for the original VRPTW (see Desaulniers et al. [9] and Baldacci et al. [2]) and the solutions of the VRPFlexTW. This table represents the total distance and the number of vehicles for the following cases: (i) the optimal/best-known solutions of the original VRPTW, (ii) the final solutions obtained by our solution procedure for the original VRPTW (VRPFlexTW with $p_i = 0$, $\forall i \in N$), (iii) the final solutions obtained by our solution procedure for the VRPFlexTW with $p_i = 0.05$, $\forall i \in N$, (iv) the final solutions obtained by our solution procedure for the VRPFlexTW with $p_i = 0.10$, $\forall i \in N$, (v) the final solutions obtained by our solution procedure for the VRPFlexTW with $p_i = 0.15$, $\forall i \in N$. These solutions are generated by setting (θ,τ) to (10^5,10^4) in the tabu search algorithm. The average values of the total distance and the number of vehicles over all problem instances are also given in Table 6. Overall, the results indicate that VRPFlexTW with a positive flexibility percentage (p_i) provides a decrease in the average number of vehicles compared to that obtained by the optimal/best-known solutions of the original VRPTW. Furthermore, the average total distance and the average number of vehicles are decreasing as p_i increases.

Results obtained for the original VRPTW (VRPFlexTW with $p_i = 0$, $\forall i \in N$) show that our solution procedure obtains good final solutions with respect to the optimal/best-known solutions. Moreover, for six problem instances (c101, c105, c106, c107, c108, c109) we obtain the optimal solutions. Since our solution procedure is effective for the original problem, we first compare the final solutions obtained for the VRPFlexTW (cases (iii),(iv) and (v)) with the final solutions obtained by our solution procedure for the VRPTW (case (ii)). We also compare the final solutions obtained for the VRPFlexTW with the optimal/best-known solutions of the original VRPTW (case (i)).

Table 6: Comparison of the original VRPTW solutions with the final solutions obtained by our solution procedure where $(\theta,\tau)=(10^5,10^4)$

	Opt. VRPTW		VRPFlexTW, p_i=0		VRPFlexTW, p_i=0.05		VRPFlexTW, p_i=0.10		VRPFlexTW, p_i=0.15	
Ins.	Dist.	#Veh.	Dist.	#Veh.	Dist.	#Veh.	Dist.	#Veh.	Dist.	#Veh.
c101	828.94	10	828.94	10	828.94	10	828.94	10	828.94	10
c102	828.94	10	1014.31	10	828.94	10	886.03	10	969.12	10
c103	828.06	10	896.88	10	888.81	10	836.45	10	934.25	11
c104	824.78	10	831.90	10	873.83	10	936.10	10	902.01	10
c105	828.94	10	828.94	10	828.94	10	884.25	10	828.94	10
c106	828.94	10	828.94	10	828.94	10	828.94	10	828.94	10
c107	828.94	10	828.94	10	828.94	10	828.94	10	828.94	10
c108	828.94	10	828.94	10	828.94	10	828.94	10	828.94	10
c109	828.94	10	828.94	10	828.94	10	828.94	10	828.94	10
r101	1642.92	20	1681.85	20	1661.10	19	1641.26	18	1685.69	18
r102	1471.75	18	1487.24	18	1469.78	17	1431.27	17	1453.48	17
r103	1213.62	14	1246.03	14	1264.75	14	1260.96	13	1207.84	14
r104	976.76	11	1016.57	11	1017.38	10	998.49	10	1012.73	10
r105	1360.12	15	1394.71	15	1373.56	15	1359.13	15	1359.70	14
r106	1239.37	13	1282.29	13	1305.30	13	1289.68	13	1209.18	13
r107	1069.09	11	1094.67	12	1075.55	12	1081.60	11	1056.33	11
r108	936.69	10	965.36	10	954.93	10	990.70	9	967.38	10
r109	1151.89	13	1159.16	12	1153.50	12	1138.13	12	1141.04	12
r110	1072.41	12	1100.18	12	1106.47	12	1056.31	11	1039.12	11
r111	1053.50	12	1078.27	12	1079.93	12	1053.35	11	1063.42	11
r112	953.44	10	974.48	10	974.32	10	992.27	10	982.57	10
rc101	1623.58	15	1669.88	16	1641.65	15	1611.63	15	1603.32	14
rc102	1461.33	14	1531.26	14	1503.75	14	1481.62	13	1461.05	13
rc103	1261.67	11	1299.83	12	1283.40	12	1247.43	11	1209.30	11
rc104	1135.48	10	1171.15	11	1143.93	10	1153.65	11	1137.11	10
rc105	1517.93	15	1563.56	15	1576.17	15	1453.08	14	1472.33	14
rc106	1376.26	12	1398.58	13	1363.55	12	1310.57	12	1297.37	12
rc107	1211.24	12	1257.69	12	1264.87	11	1207.94	11	1212.36	11
rc108	1117.53	11	1149.89	11	1115.90	10	1113.87	11	1129.53	10
Avg.	1113.86	12.03	1146.18	12.17	1134.31	11.90	1122.77	11.66	1119.99	11.62

5.3.1. VRPFlexTW versus VRPTW with the solutions obtained by our solution procedure

Results obtained with $p_i = 0.05$, $\forall i \in N$ show that for six problem instances (r101, r102, rc101, rc104, rc106, rc108) VRPFlexTW reduces both the total distance and the number of vehicles, compared to those obtained by our solution procedure for the original VRPTW where time windows are defined as hard time windows. For nine problem instance (c102, c103, r105, r107, r108, r109, r112, rc102, rc103), VRPFlexTW provides a reduction in the total distance with the same number of vehicles as the one given by our solutions obtained for the original VRPTW. For two problem instances (r104, rc107), VRPFlexTW yields fewer vehicles; however, this brings an increase in the total distance. For six problem instances (c101, c105, c106, c107, c108, c109), VRPFlexTW obtains the same solutions as the ones found by our solution procedure for the original VRPTW. For the remaining problem instances, VRPFlexTW results in an increase in the total distance with the same number of vehicles.

Results obtained with $p_i = 0.10$, $\forall i \in N$ show that for 12 problem instances (r101, r102, r104, r107, r110, r111, rc101, rc102, rc103, rc105, rc106, rc107) VRPFlexTW reduces both the total distance and the number of vehicles, compared to those obtained by our solution procedure for the original VRPTW where time windows are defined as hard time windows. For six problem instances (c102, c103, r105, r109, rc104, rc108), VRPFlexTW provides a reduction in the total distance with the same number of vehicles as the one given by our solutions obtained for the original VRPTW. For two problem instances (r103, r108), VRPFlexTW yields fewer vehicles; however, this brings an increase in the total distance. For five problem instances (c101, c106, c107, c108, c109), VRPFlexTW obtains the same solutions as the ones found by our solution procedure for the original VRPTW. For the remaining problem instances, VRPFlexTW results in an increase in the total distance with the same number of vehicles.

Results obtained with $p_i = 0.15$, $\forall i \in N$ show that for 14 problem instances (r102, r104, r105, r107, r110, r111, rc101, rc102, rc103, rc104, rc105, rc106, rc107, rc108) VRPFlexTW reduces both the total distance and the number of vehicles, compared to those obtained by our solution procedure for the original VRPTW where time windows are defined as hard time windows. For four problem instances (c102, r103, r106, r109), VRPFlexTW provides a reduction in the total distance with the same number of vehicles as the one

given by our solutions obtained for the original VRPTW. For one problem instance (r101), VRPFlexTW yields fewer vehicles; however, this brings an increase in the total distance. For six problem instances (c101, c105, c106, c107, c108, c109), VRPFlexTW obtains the same solutions as the ones found by our solution procedure for the original VRPTW. For the remaining problem instances, VRPFlexTW results in an increase either in the total distance, or both in the total distance and the number of vehicles.

5.3.2. VRPFlexTW versus VRPTW with the optimal/best-known solutions

Note that in this subsection, we refer to the optimal/best-known solutions of the original VRPTW as the optimal VRPTW. Results obtained with $p_i = 0.05$, $\forall i \in N$ show that for two problem instances (r102, rc108) VRPFlexTW reduces both the total distance and the number of vehicles, compared to those obtained by the optimal VRPTW where time windows are defined as hard time windows. For one problem instance (rc106), VRPFlexTW provides a reduction in the total distance with the same number of vehicles as the one given by the optimal VRPTW. For four problem instances (r101, r104, r109, rc107), VRPFlexTW yields fewer vehicles; however, this brings an increase in the total distance. For seven problem instances (c101, c102, c105, c106, c107, c108, c109), VRPFlexTW obtains the same solutions as the optimal VRPTW. For the remaining problem instances, VRPFlexTW results in an increase either in the total distance, or both in the total distance and the number of vehicles.

Results obtained with $p_i = 0.10$, $\forall i \in N$ show that for seven problem instances (r101, r102, r109, r110, r111, rc105, rc107) VRPFlexTW reduces both the total distance and the number of vehicles, compared to those obtained by the optimal VRPTW where time windows are defined as hard time windows. For five problem instances (r105, rc101, rc103, rc106, rc108), VRPFlexTW provides a reduction in the total distance with the same number of vehicles as the one given by the optimal VRPTW. For four problem instances (r103, r104, r108, rc102), VRPFlexTW yields fewer vehicles; however, this brings an increase in the total distance. For five problem instances (c101, c106, c107, c108, c109), VRPFlexTW obtains the same solutions as the optimal VRPTW. For the remaining problem instances, VRPFlexTW results in an increase either in the total distance, or both in the total distance and the number of vehicles.

Results obtained with $p_i = 0.15$, $\forall i \in N$ show that for seven problem instances (r102, r105, r109, r110, rc101, rc102, rc105) VRPFlexTW reduces

both the total distance and the number of vehicles, compared to those obtained by the optimal VRPTW where time windows are defined as hard time windows. For five problem instances (r103, r106, r107, rc103, rc106), VRPFlexTW provides a reduction in the total distance with the same number of vehicles as the one given by the optimal VRPTW. For five problem instance (r101, r104, r111, rc107, rc108), VRPFlexTW yields fewer vehicles; however, this brings an increase in the total distance. For six problem instances (c101, c105, c106, c107, c108, c109), VRPFlexTW obtains the same solutions as the optimal VRPTW. For the remaining problem instances, VRPFlexTW results in an increase either in the total distance, or both in the total distance and the number of vehicles.

6. Conclusions

In this paper, we introduce the VRPFlexTW which enables to serve customers outside their original time boundaries with respect to a given tolerance. Compared to the VRPTW, the VRPFlexTW permits fixed deviations from customer time windows at a cost. Furthermore, when compared to the VRPSTW, the VRPFlexTW operates on a far more restricted solution space.

Our solution procedure comprises three main components: initialization, routing and scheduling. The time-oriented nearest neighbor heuristic is used in the initialization component. The routing component is handled via a tabu search algorithm, while the scheduling component is performed by solving an LP model. We validate our solution algorithm on benchmark instances and test the performance of the solution procedure with various stopping criteria values. Furthermore, we compare the solutions of the VRPTW with those of the VRPFlexTW. In many instances, we observe that the VRPFlexTW results in operational gains when compared to the VRPTW. These gains are achieved by a reduction in the total distance traveled or by a reduction in the number of vehicles used or by a reduction both in the total distance and in the number of vehicles.

We model a practical problem and develop an efficient solution framework to handle it. Our solution approach can effectively be used by carrier companies trying to assess the added value of allowing a certain extent of customer service flexibility. Further research may focus on handling uncertainties in travel times and on exploring more complex penalty functions.

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